

HAT-P-37b

R-2294

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_221105 · created 2026-08-17T01:09:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459888.646 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.045
Transit depth (Rp/Rs)^2	2.74 +/- 1.49 [%]
Orbital Inclination (inc)	83.17 +/- 2.45
Airmass coefficient 1 (a1)	0.998 +/- 0.027
Airmass coefficient 2 (a2)	0.0037 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	2.72 %
Best Comparison Star	#3 - [146, 439]
Optimal Aperture	3.7
Optimal Annulus	11.47
Transit Duration (day)	0.047 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.045 vs 0.1394 (deviation 0.42 σ)
Duration fit vs published	0.047 d vs 0.0983 d (deviation 1.01 σ)
Mid-time O-C	-21.2 min
Depth SNR	2.7
Residual scatter	2.72 %

Light curve

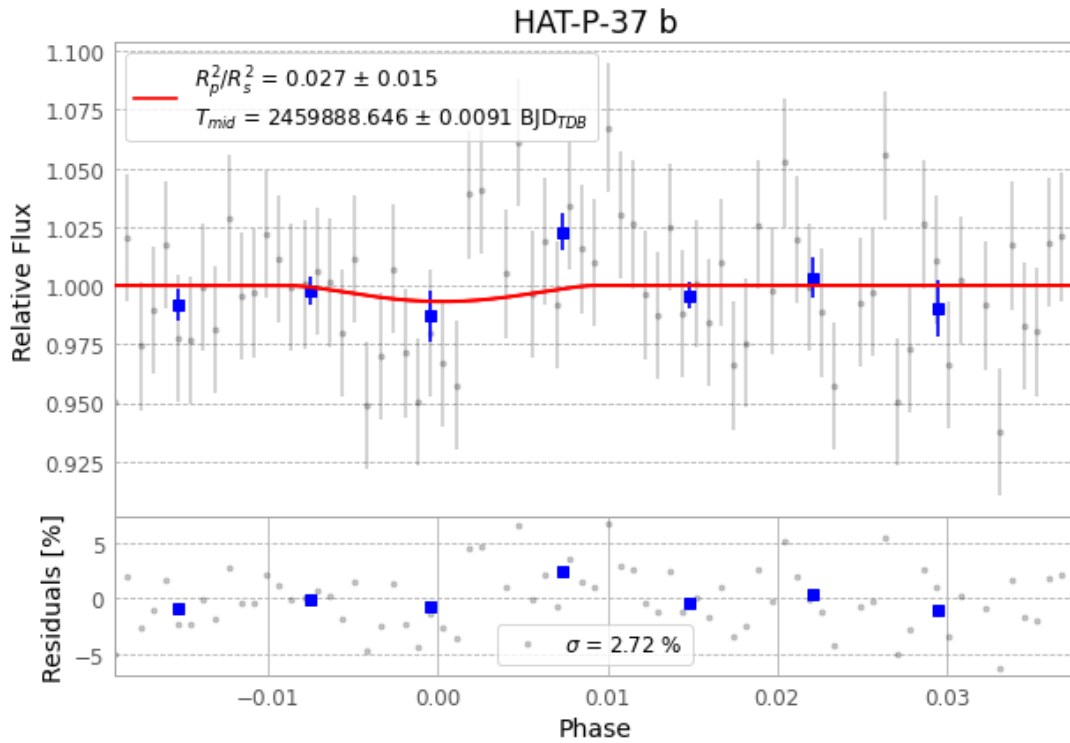


Figure 1 — FinalLightCurve_HAT-P-37 b_2022-11-04.png (SHA-256 b53db65d06165edf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [429, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d6b1263f42e3db9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

78 FITS frames (51 MB), HATP-37221105021216.FITS → HATP-37221105060315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-221105.log (e41f40c2e1d7...), FinalLightCurve_HAT-P-37 b_2022-11-04.png (b53db65d0616...), FinalLightCurve_HAT-P-37 b_2022-11-04.csv (3a4bf86cb6aa...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 01:09Z.